

Deva Anand

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Education

University of Massachusetts Amherst

Master of Science in Computer Science

September 2025 – May 2027 (Expected)

Amherst, MA

Coursework: Advanced Machine Learning, Advanced NLP, Optimization in CS, Systems for Data Science, Performance Engineering

Vels Institute of Science, Technology and Advanced Studies

Bachelor of Engineering in Computer Science and Engineering

August 2019 – May 2023

Chennai, India

Publication

- “Alzheimer’s Disease Classification using Transfer Learning,” **IEEE CONIT 2023** (3rd International Conference on Intelligent Technologies), applied deep transfer learning on neuroimaging data for medical image classification (first author).

Skills

Languages

Python, C++, TypeScript, JavaScript, SQL, Bash

ML & Deep Learning

PyTorch, JAX, HuggingFace Transformers, training loops, custom optimizers, autodiff from scratch, normalizing flows, diffusion, transfer learning, NumPy, scikit-learn

LLM & Agentic Systems

LLM agents, tool use, verifiable evaluation, holdout-protected fitness, RAG, RRF, cross-encoder reranking, MMR, knapsack token budgeting, Ollama, llama-index, spaCy, sentence-transformers

Backend & Experiment

FastAPI, Node.js, REST APIs, async services, unit testing, batch experiment harnesses, Linux profiling, Apache Spark /

Infra

PySpark, MongoDB, PostgreSQL, SQLite

Cloud & DevOps

AWS, GCP, Azure, Docker, Git, GitHub Actions, CI/CD

Research & Projects

Generative Models & Deep Learning From Scratch

github.com/Deva-1903/UMASS_CICS_689

JAX, PyTorch, NumPy | UMass CS 689 Advanced ML

- Implemented **RealNVP normalizing flows** and **DDPM diffusion models** from scratch in JAX, including custom forward/reverse processes, training loops, custom optimizers, and likelihood + sample-quality diagnostics on image benchmarks.
- Built and benchmarked **8 deep architectures** (CNN / ResNet variants, regularized models) on CIFAR-10; implemented **reverse-mode automatic differentiation** over matrix-valued operations from scratch to validate gradient correctness against framework autodiff.

KG2RAG-Enhanced - Multi-Hop QA with Hierarchical Decomposition

github.com/Deva-1903/KG2RAG-685-NLP

Python, Ollama, sentence-transformers, llama-index, spaCy, PyTorch | UMass CS 685 Advanced NLP

- Extended KG2RAG with **hierarchical sub-question decomposition** and multi-view seed retrieval on HotpotQA: per-view passage retrieval fused via Reciprocal Rank Fusion (RRF), cross-encoder reranking, and Maximal Marginal Relevance (MMR).
- Replaced heuristic top-M selection with a **0–1 knapsack token-budget** formulation (value = relevance × coverage) for **context compaction** under bounded context windows; ran **100–500-question batch experiments** via an Ollama + LLaMA-3 inference pipeline with token and latency logging.

AutoEval - Self-Improving Agent with Verifiable Acceptance

github.com/Deva-1903/AutoEval

Python, IR Evaluation, Git Automation

- Designed a **self-improvement agent** for retrieval evaluation that proposes new queries and labels, scores them against multiple reference systems, and accepts only updates that improve discriminative power on a **holdout-heavy verifiable fitness objective**.
- Added anti-Goodharting guardrails (**schema checks, duplicate rejection, read-only holdout protection**) so structured feedback can refine the eval suite without overfitting, and committed accepted updates back to Git with report cards for an auditable, reproducible workflow.

Experience

Wysa

July 2023 – July 2025

Backend Engineer

Bengaluru, India

- Architected a full-stack **ML training-data annotation platform** (React, Node.js, MongoDB) used daily by the AI team to label data for production NLP/clinical models; replaced spreadsheet workflows with secure role-based APIs that accelerated iteration on downstream model training by **100+ hours/month**.
- Engineered multi-tenant NHS eTriage backend (Node.js, MongoDB) for **8+ UK clients** processing **10,000+ monthly submissions**; owned reliability and remediated **20+ VAPT findings**.

Extracurriculars

- Selected for **GirlScript Summer of Code (GSSoC) 2023**; contributed to multiple open-source projects (Linkfree, Freehit, ProjectsHut).
- Tutored **30+** underprivileged students in programming at Sayur, non-profit in South Tamil Nadu.